

SeasonStereo: Robust Dense Stereo Matching for Multi-Date Satellite Imagery via Generative AI

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Motivation

Satellite stereo matching assumes near-simultaneous image pairs, but real archives contain multi-date images with strong seasonal and illumination changes. Existing methods fail under these diachronic conditions, and training robust models requires expensive LiDAR supervision.



Contributions

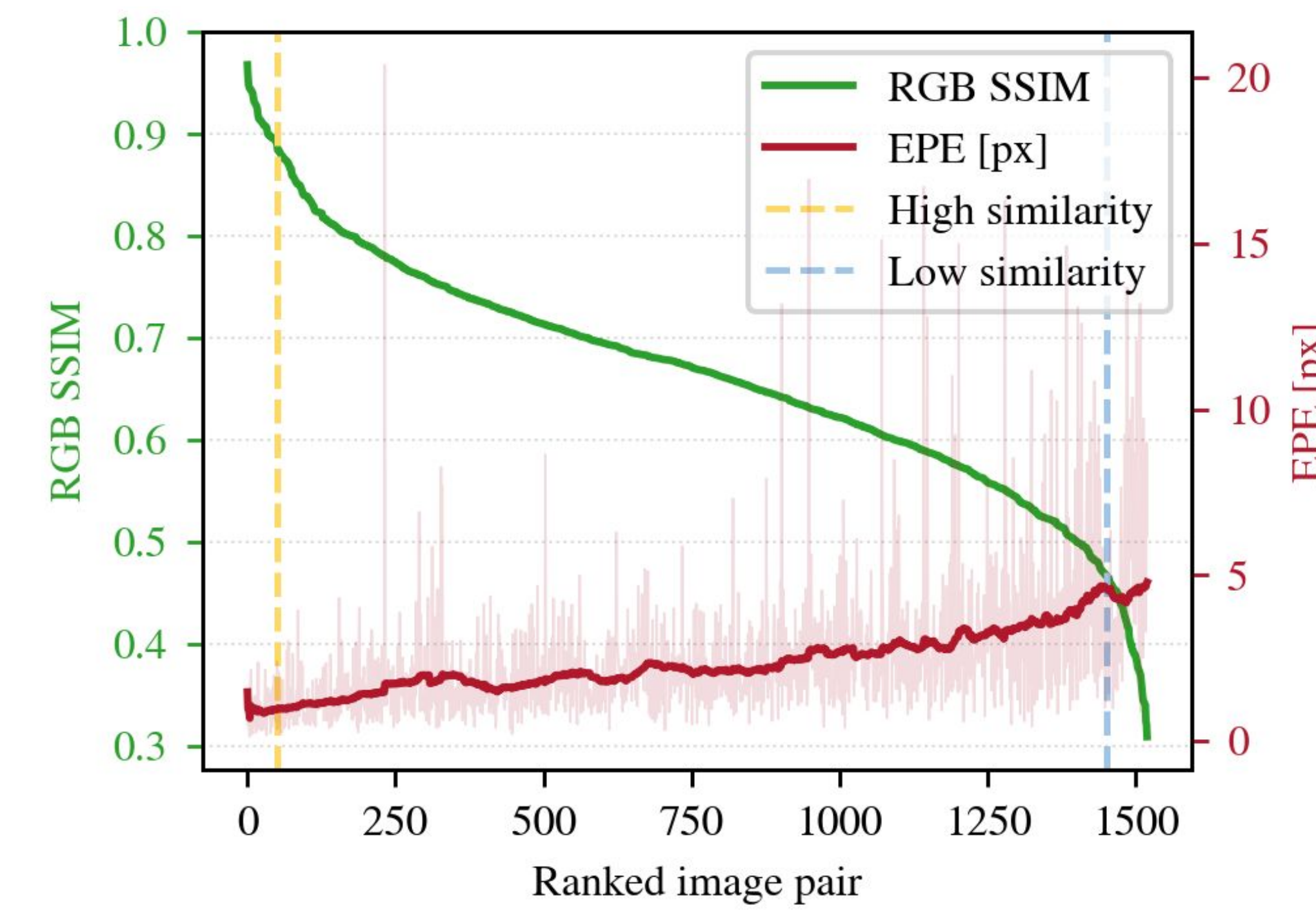
We propose SeasonStereo, a scalable framework featuring:

1. A framework for dense stereo matching of multi-date satellite images using synthetic seasonal appearance, no LiDAR supervision or real diachronic products needed.
2. Multi-term training objective combining zero-shot foundation model priors with photometric and smoothness terms for sharper building contours.
3. Large-scale dataset of 21,775 aligned real and synthetic stereo pairs in synchronic and diachronic settings.

Method

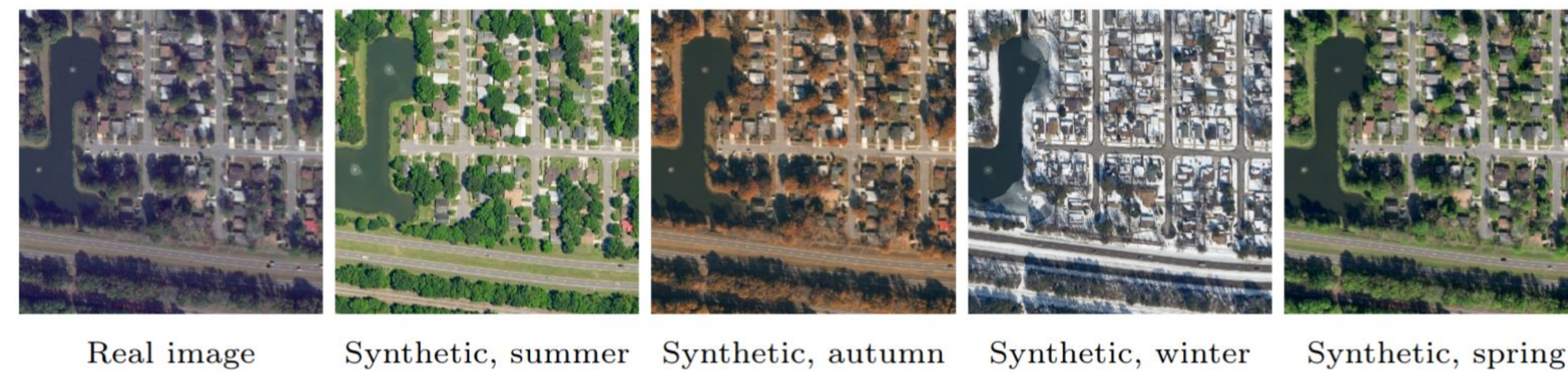
1 Data Curation

From 1,565 candidate synchronic pairs, we select 871 highly consistent pairs via multi-metric consensus: combined RGB-SSIM photometric similarity and DINOv3 feature cosine similarity. Pairs with low consistency are discarded, as lower similarity correlates with higher MonSter++ disparity error.



2 Seasonal synthesis

For each image, Nano Banana Pro generates four geometry-preserving seasonal variants from text prompts. Combined with originals, this yields 25 left-right combinations per pair, expanding training from 871 to 21,775 pairs.



3 Training

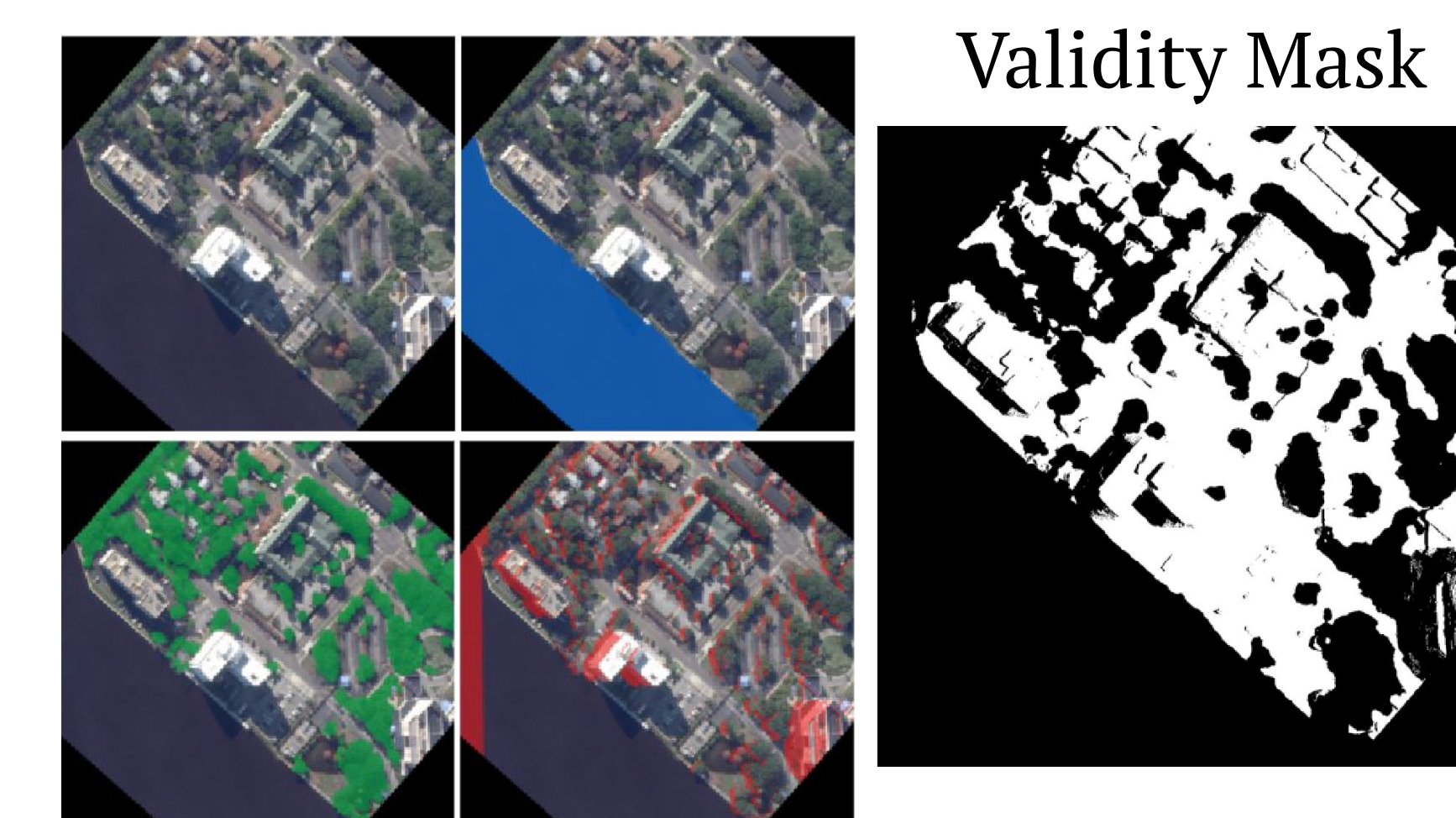
We fine-tune MonSter++ (stereo backbone + update modules; monocular branch frozen) using a multi-term loss:

$$\mathcal{L}^{(k)} = \underbrace{\lambda_{\text{disp}} \mathcal{L}_{\text{disp}}(\hat{D}^{(k)})}_{[1]} + \underbrace{\lambda_{\text{photo}} \mathcal{L}_{\text{photo}}(\hat{D}^{(k)})}_{[2]} + \underbrace{\lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}(\hat{D}^{(k)})}_{[3]}$$

$$[1] \mathcal{L}_{\text{disp}}(\hat{D}^{(k)}) = \frac{1}{|\mathcal{V}|} \sum_{\mathbf{x} \in \mathcal{V}} \left| \hat{D}^{(k)}(\mathbf{x}) - \tilde{D}(\mathbf{x}) \right|$$

$$[2] \mathcal{L}_{\text{photo}}(\hat{D}^{(k)}) = \frac{1}{|\mathcal{V} \cap \mathcal{V}_B|} \sum_{\mathbf{x} \in \mathcal{V} \cap \mathcal{V}_B} \ell_{\text{photo}} \left(I_L^{\text{real}}(\mathbf{x}), \text{warp}(I_R^{\text{real}}, \hat{D}^{(k)})(\mathbf{x}) \right)$$

$$[3] \mathcal{L}_{\text{smooth}}(\hat{D}^{(k)}) = \frac{1}{|\mathcal{V}|} \sum_{\mathbf{x} \in \mathcal{V}} \left(\left| \partial_x \hat{D}^{(k)}(\mathbf{x}) \right| e^{-|\partial_x I_L(\mathbf{x})|} + \left| \partial_y \hat{D}^{(k)}(\mathbf{x}) \right| e^{-|\partial_y I_L(\mathbf{x})|} \right)$$



Quantitative Results

Method	JAX	Buenos Aires	OMA Sync.	OMA Diach.	Avg. MAE
MonSter	1.70	2.32	0.87	1.59	1.62
MonSter++	1.53	2.38	0.87	2.93	1.93
Diachronic Stereo	1.12	1.54	0.75	0.82	1.06
(a) L_LiDAR	1.17	1.50	0.74	0.80	1.05
(b) L_disp only	1.20	1.48	0.76	0.79	1.06
(c) L_disp + L_photo	1.18	1.58	0.77	0.84	1.09
SeasonStereo (ours)	1.15	1.52	0.75	0.79	1.05

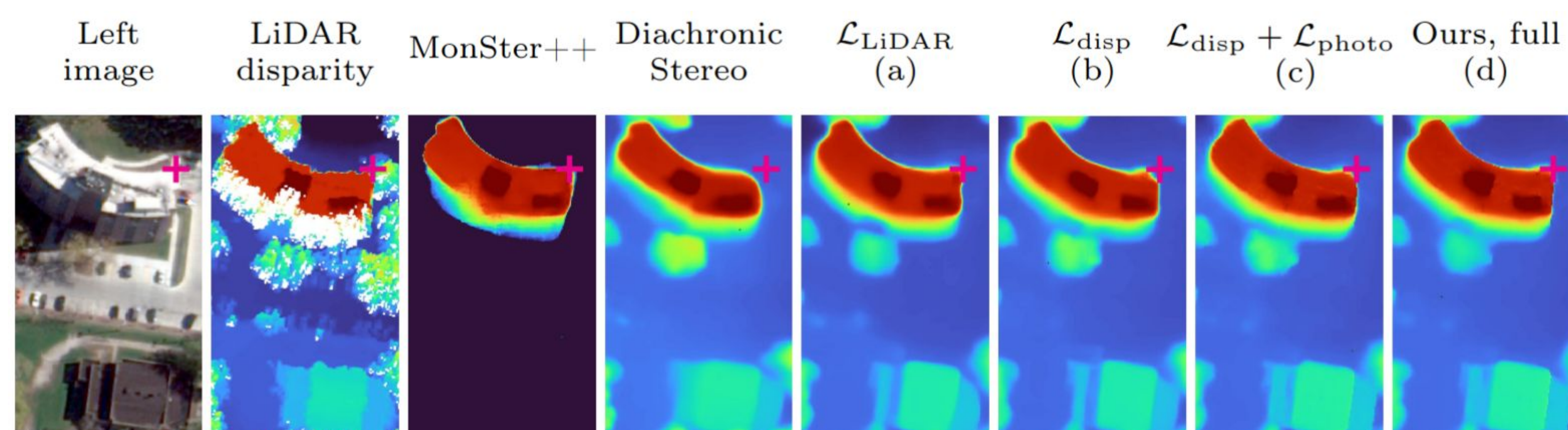
Altitude MAE (m) across test sets, baselines and SeasonStereo loss ablation variants.

Training data	# Pairs	OMA Diach.	Avg. MAE
S1: real only	871	1.04	1.18
S2: +synch. synth.	871 + 3,483	0.88	1.10
S3: +mixed synth.	871 + 3,483	0.81	1.07

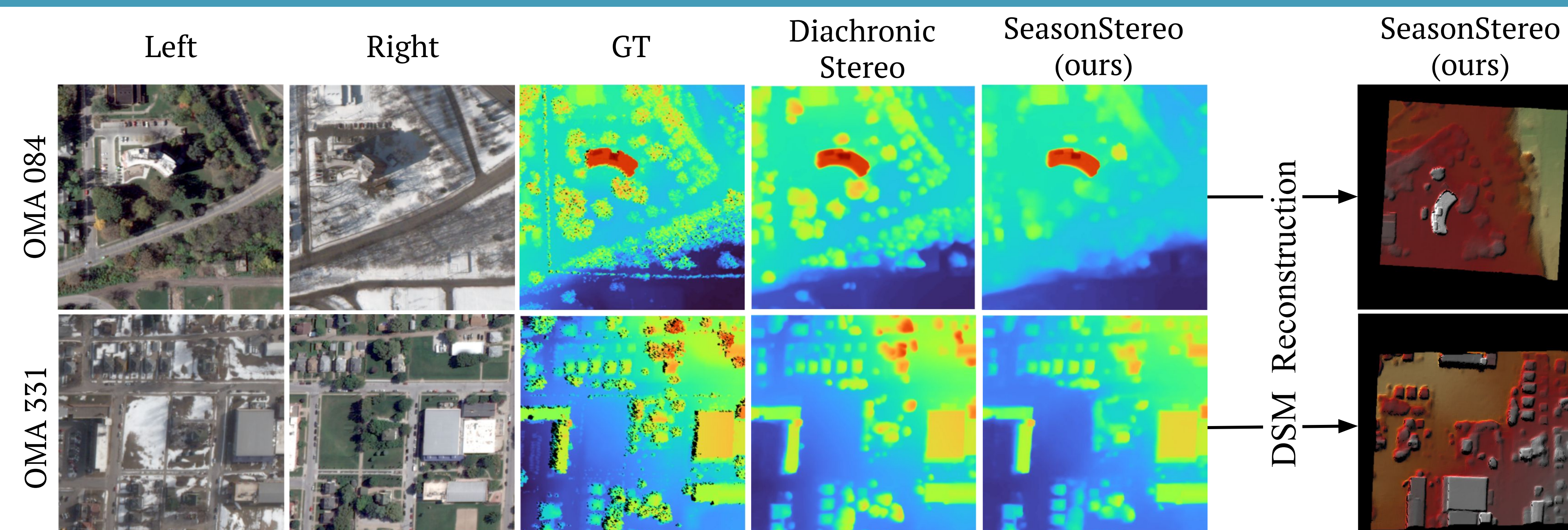
Impact of synthetic diachronic training data on altitude MAE (m).

SeasonStereo matches LiDAR-supervised accuracy (1.05 m vs. 1.06 m avg. MAE) with sharper building contours, without any LiDAR labels or real diachronic data. Synthetic diachronic augmentation is key: on the hardest test set (Omaha Diachronic), MAE drops from 1.04 m to 0.81 m. These results suggest that generative appearance synthesis, coupled with foundation model priors, is a scalable alternative to costly ground-truth acquisition.

Qualitative Results



Ablation study on OMA 084. Full model (d) recovers the sharpest building edges.



Check it yourself!

<https://multimedia-eurecat.github.io/SeasonStereo/>